

02_analysis part

September 27, 2026

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[1]: import pickle

import matplotlib.pyplot as plt
import torch
from sklearn.metrics import ConfusionMatrixDisplay
from tqdm import tqdm

from sdoml_task1.config import PROJECT_DIR
from sdoml_task1.modeling.model import Net

[3]: save_dir = PROJECT_DIR / "models"

with open(save_dir / "history.pkl", "rb") as f:
    history = pickle.load(f)

total_loss_train = history["total_loss_train"]
total_accuracy_train = history["total_accuracy_train"]
total_loss_valid = history["total_loss_valid"]
total_accuracy_valid = history["total_accuracy_valid"]
all_confusion_matrices = history["all_confusion_matrices"]
total_loss_train = history["total_loss_train"]
input_dim = history["input_dim"]

# This loads the model
# net = Net(input_dim=input_dim)
# net.load_state_dict(torch.load(save_dir / "model.pt", map_location="cpu"))
# net.eval()

# torch.save(net.state_dict(), save_dir / "model.pt")

[5]: for i, cm in enumerate(tqdm(all_confusion_matrices, desc="Generating confusion_
    ↪matrices")):
    disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=range(10))
    disp.plot(cmap='Blues')
    plt.title(f"Confusion Matrix - Epoch {i+1}")
    plt.savefig(f"../assets/model_analysis/Confussion_Matrix_Epoch{i+1}.png",
    ↪dpi=150)
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plt.close()
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Generating confusion matrices: 100%| | 40/40 [00:12<00:00, 3.21it/s]

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[8]: fig, axs = plt.subplots(2, 2, figsize=(12, 8))
      ax1, ax2, ax3, ax4 = axs.flatten()

      ax1.plot(total_loss_train)
      ax1.set_xlabel('Epoch')
      ax1.set_ylabel('Loss')
      ax1.set_title('Training Loss')
      ax1.grid()

      ax2.plot(total_accuracy_train)
      ax2.set_xlabel('Epoch')
      ax2.set_ylabel('Accuracy')
      ax2.set_title('Training Accuracy')
      ax2.grid()

      ax3.plot(total_loss_valid)
      ax3.set_xlabel('Epoch')
      ax3.set_ylabel('Loss')
      ax3.set_title('Validation Loss')
      ax3.grid()

      ax4.plot(total_accuracy_valid)
      ax4.set_xlabel('Epoch')
      ax4.set_ylabel('Accuracy')
      ax4.set_title('Validation Accuracy')
      ax4.grid()

      plt.tight_layout()
      plt.savefig(f"../assets/model_analysis/loss_accuracy_comparison.png", dpi=150)
      plt.close()
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